

Deep Reinforcement Learning

Deep Q Learning

Prof. Dr. Sebastian Peitz

Chair of Safe Autonomous Systems, TU Dortmund

Summer term 2026

Content

Content

- Recap:
 - Tabular SARSA & Q -learning
 - Value function approximation with NNs
- On-policy control with gradients and semi-gradients
- Deep Q networks (DQN)
 - Extensions & modifications
- Least squares policy iteration

Where are we?

Chapter	Topic	Content
	Basics & tabular methods	
1-5	Bandits, MDPs, Dynamic Programming, Monte Carlo, TD Learning	RL basics in finite dimensions
	Deep-learning-based methods	
6	Brief introduction to deep learning	The basics for what comes next
7	Value function approximation	Value estimation with function approximation
8	Deep Q -learning	Q -learning with neural networks
9	Policy gradients	
10	Actor-critic algorithms	
11	Advanced algorithms (Part I): From policy gradient to PPO	
12	Advanced algorithms (Part II): From Q -learning to Soft Actor-Critic	
13	Exploration	
	Model-Based Control	
	Advanced Topics	

Chapter	Topic	Content
---------	-------	---------

Lecture contents

Recap: Tabular SARSA and Q -learning

Tabular SARSA and Q -learning

Algorithm: SARSA (**On-policy**).

initialize

- $Q(s, a)$ arbitrarily for $s \in \mathcal{S}, a \in \mathcal{A}$
- $Q(\text{terminal}, \cdot) = 0$
- $\pi = \epsilon\text{-greedy}(Q)$

for $k = 1, 2, \dots, K$ episodes:

Initialize $s_t \leftarrow s_0, t \leftarrow 0$

while s_t is not terminal:

Take action $a_t \sim \pi(s_t)$ and observe (r_t, s_{t+1})

Select $a_{t+1} \sim \pi(s_{t+1})$

Update Q given $(s_t, a_t, r_t, s_{t+1}, a_{t+1})$:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha [r_t + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t)]$$

Update policy: $\pi = \epsilon\text{-greedy}(Q)$

$t \leftarrow t + 1$

Algorithm: Q -learning (**Off-policy**).

initialize

- $Q(s, a)$ arbitrarily for $s \in \mathcal{S}, a \in \mathcal{A}$
- $Q(\text{terminal}, \cdot) = 0$
- $\pi = \epsilon\text{-greedy}(Q)$

for $k = 1, 2, \dots, K$ episodes:

Initialize $s_t \leftarrow s_0, t \leftarrow 0$

while s_t is not terminal:

Take action $a_t \sim \pi(s_t)$ and observe (r_t, s_{t+1})

~~Select $a_{t+1} \sim \pi(s_{t+1})$~~

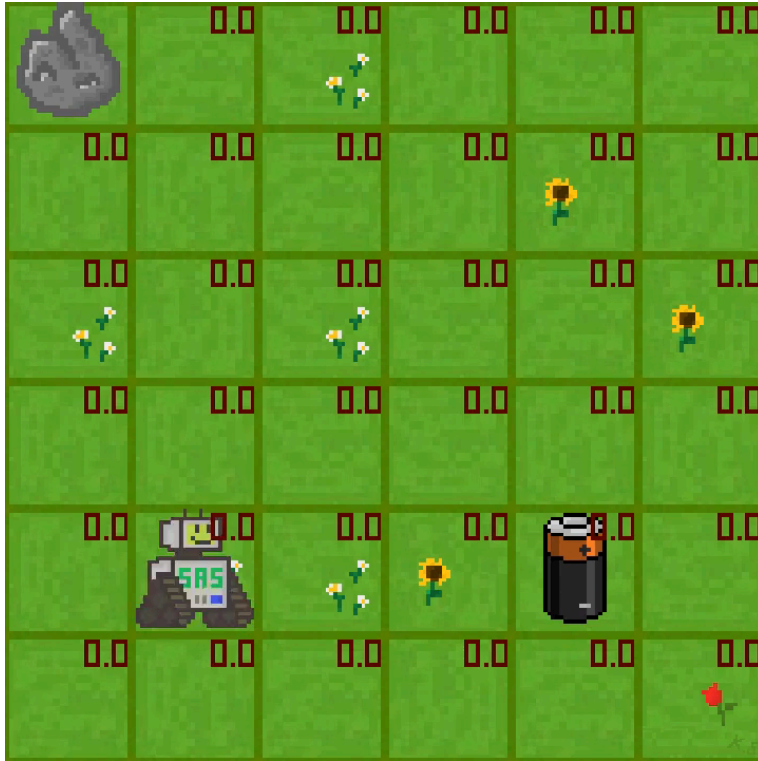
Update Q given (s_t, a_t, r_t, s_{t+1}) :

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[r_t + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t) \right]$$

Update policy $\pi = \epsilon\text{-greedy}(Q)$

$t \leftarrow t + 1$

Tabular SARSA and Q -learning: Gridworld



On-policy: SARSA



Off-policy: Q -learning

- Discount: $\gamma = 0.9$
- Exploration: $\epsilon = 0.2$
- Step size: $\alpha = 0.5$

Recap: Value function approximation

- We want to **approximate the value function by parametric model** with trainable parameters $\theta \in \mathbb{R}^d$, i.e., $V_\theta(s) \approx V^\pi(s)$.
- For training, we need to define a **prediction objective**:

$$L(\theta) = \sum_{k=1}^N \left(V^\pi(s_k) - V_\theta(s_k) \right)^2 \approx \int_{\mathcal{S}} \mu(s) \left(V^\pi(s) - V_\theta(s) \right)^2 ds = \overline{VE}(\theta). \quad (1)$$

- **Incremental weight updates** via
 - gradients, e.g., in the Monte Carlo setting:

$$\theta \leftarrow \theta + \alpha [g - V_\theta(s)] \nabla_\theta V_\theta(s). \quad (2)$$

- semi-gradients in the bootstrapping case: we are not differentiating the target, which also depends on V_θ . For TD learning, we have

$$\theta \leftarrow \theta + \alpha [r + \gamma V_\theta(s') - V_\theta(s)] \nabla_\theta V_\theta(s). \quad (3)$$

- **Batch updates** using $\mathcal{B} = \{(s_i, r_i, s'_i)\}_{i=1}^b \subset \mathcal{D}$ instead of a single sample as in SGD. For TD updates, we thus get

$$\theta \leftarrow \theta + \frac{\alpha}{b} \sum_{(s_i, r_i, s'_i) \in \mathcal{B}} [r_i + \gamma V_\theta(s'_i) - V_\theta(s_i)] \nabla_\theta V_\theta(s_i). \quad (4)$$

- For **linear models** (i.e., $V_\theta(s) = \theta^\top \psi(s) = \sum_{k=1}^d \theta_k \psi_k(s)$), training can be done in one step via linear regression:

$$\theta^* = (\Psi^\top \Psi)^{-1} \Psi^\top y = \Psi^\dagger y \quad \Rightarrow \quad V_{\theta^*}(s) = \theta^{*\top} \psi(s) = \sum_{k=1}^d \theta_k^* \psi_k(s). \quad (5)$$

On-policy control with gradients and semi-gradients

Value-based control: Q -function approximation

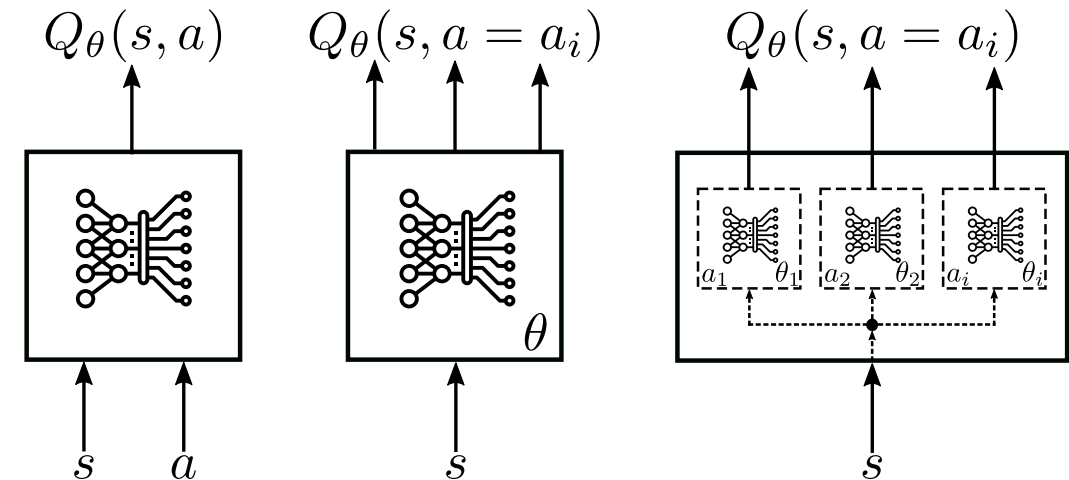
Today's focus: **value-based control** $\Rightarrow$ How to transfer tabular methods to the function approximation case.

- The states are continuous, but the actions are (usually) still discrete.
- Central object of interest: The Q function that we wish to approximate with a parametric model (e.g., a neural net)

$$Q_{\theta}(s, a) \approx Q^{\pi}(s, a)$$

- We can then use this in the well-known generalized policy iteration (GPI) scheme to identify optimal policies:

$$Q_{\theta}(s, a) \approx Q^*(s, a) \quad \Rightarrow \quad \pi^*(s) = \arg \max_{a \in \mathcal{A}} Q_{\theta}(s, a).$$



Types of architectures (Abdelwanis et al. 2026)

Gradient-based action value learning

The transfer from value function approximation is straightforward!

- Adaptation of (1) to get $L(\theta)$

$$L(\theta) = \sum_{k=1}^N \left(Q^\pi(s_k, a_k) - Q_\theta(s_k, a_k) \right)^2. \quad (6)$$

- Incremental update using (semi-)gradients:

$$\theta \leftarrow \theta + \alpha [Q^\pi(s, a) - Q_\theta(s, a)] \nabla_\theta Q_\theta(s, a). \quad (7)$$

- Depending on the control approach, the true target $Q^\pi(s, a)$ is approximated by:
 - Monte Carlo: full episodic return

$$Q^\pi(s, a) \approx g,$$

- SARSA: one-step bootstrapped estimate

$$Q^\pi(s, a) \approx r + \gamma Q_\theta(s', a'),$$

Algorithm: Gradient MC control

Algorithm: Gradient Monte Carlo algorithm for estimating Q^π / Q^*

Input:

- a differentiable, parameter-dependent function $Q_\theta : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$
- learning rate α
- **Prediction case:** a policy π
- **Improvement case:** ϵ defining the ϵ -greedy policy update based on Q_θ

Initialize: Value function weights $\theta \in \mathbb{R}^d$ arbitrarily

for $k = 1, 2, \dots, K$ episodes:

Generate a sequence following π (or ϵ -greedy(Q_θ)):

$$((s_0, a_0, r_0), (s_1, a_1, r_1), \dots, (s_{T_k-1}, a_{T_k-1}, r_{T_k-1}))$$

Calculate the every-visit returns g_t

for $t = 0, 1, \dots, T - 1$:

$$\theta \leftarrow \theta + \alpha [g_t - Q_\theta(s_t, a_t)] \nabla_\theta Q_\theta(s_t, a_t)$$

- Direct transfer from tabular case to function approximation.
- Update target becomes the sampled return $Q^\pi(s_t, a_t) \approx g_t$.
- If operating ϵ -greedy on Q_θ : baseline policy (given by $\theta^{(0)}$) must terminate the episode!

Algorithm: Semi-gradient SARSA

Algorithm: Semi-gradient SARSA for estimating Q^π / Q^*

Input:

- a differentiable, parameter-dependent function $Q_\theta : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$
- learning rate α
- **Prediction case:** a policy π
- **Improvement case:** ϵ defining the ϵ -greedy policy update based on Q_θ

Initialize: Value function weights $\theta \in \mathbb{R}^d$ arbitrarily

for $k = 1, 2, \dots, K$ episodes:

Initialize s_0

for $t = 0, 1, \dots, T - 1$:

Obtain $a_t \sim \pi(\cdot | s_t)$ (or $a_t \sim \epsilon\text{-greedy}(Q_\theta(s_t, \cdot))$)

Observe r_t and s_{t+1}

if s_{t+1} is terminal then

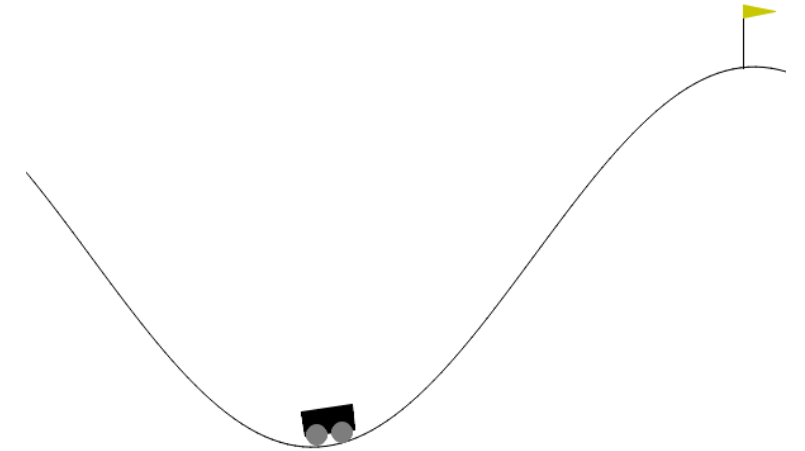
$\theta \leftarrow \theta + \alpha[r_t - Q_\theta(s_t, a_t)]\nabla_\theta Q_\theta(s_t, a_t)$

Go to next episode $k + 1$

Choose $a_{t+1} \sim \pi(\cdot | s_{t+1})$ (or $a_{t+1} \sim \epsilon\text{-greedy}(Q_\theta(s_{t+1}, \cdot))$)
 $\theta \leftarrow \theta + \alpha[r_t + \gamma Q_\theta(s_{t+1}, a_{t+1}) - Q_\theta(s_t, a_t)] \nabla_\theta Q_\theta(s_t, a_t)$

Example: Mountain car (1)

- Continuous state $s \in \mathcal{S} = [x_{\min}, x_{\max}] \times [v_{\min}, v_{\max}]$ (minimal and maximal position x and velocity v).
- Single, discrete action $a \in \mathcal{A} = \{-1, 0, 1\}$.
- $r = -1$ (goal is to terminate episode as quick as possible).
- Episode terminates when car reaches the flag (or max steps).
- Simplified longitudinal car physics with state constraints.
- Position initialized randomly within valley, zero initial velocity.
- Car is underpowered and requires swing-up.

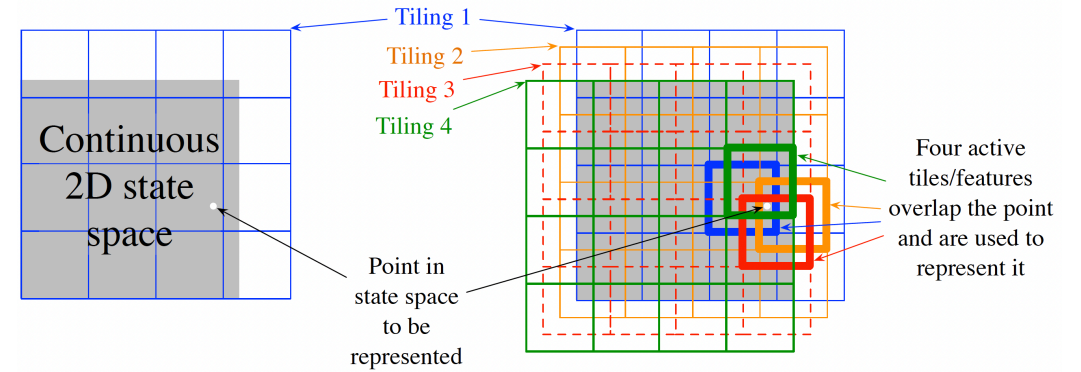


Mountain car (Sutton and Barto 1998, Gymnasium)

Approximation: Linear model with *tile coding* features $x_i(s, a)$, where

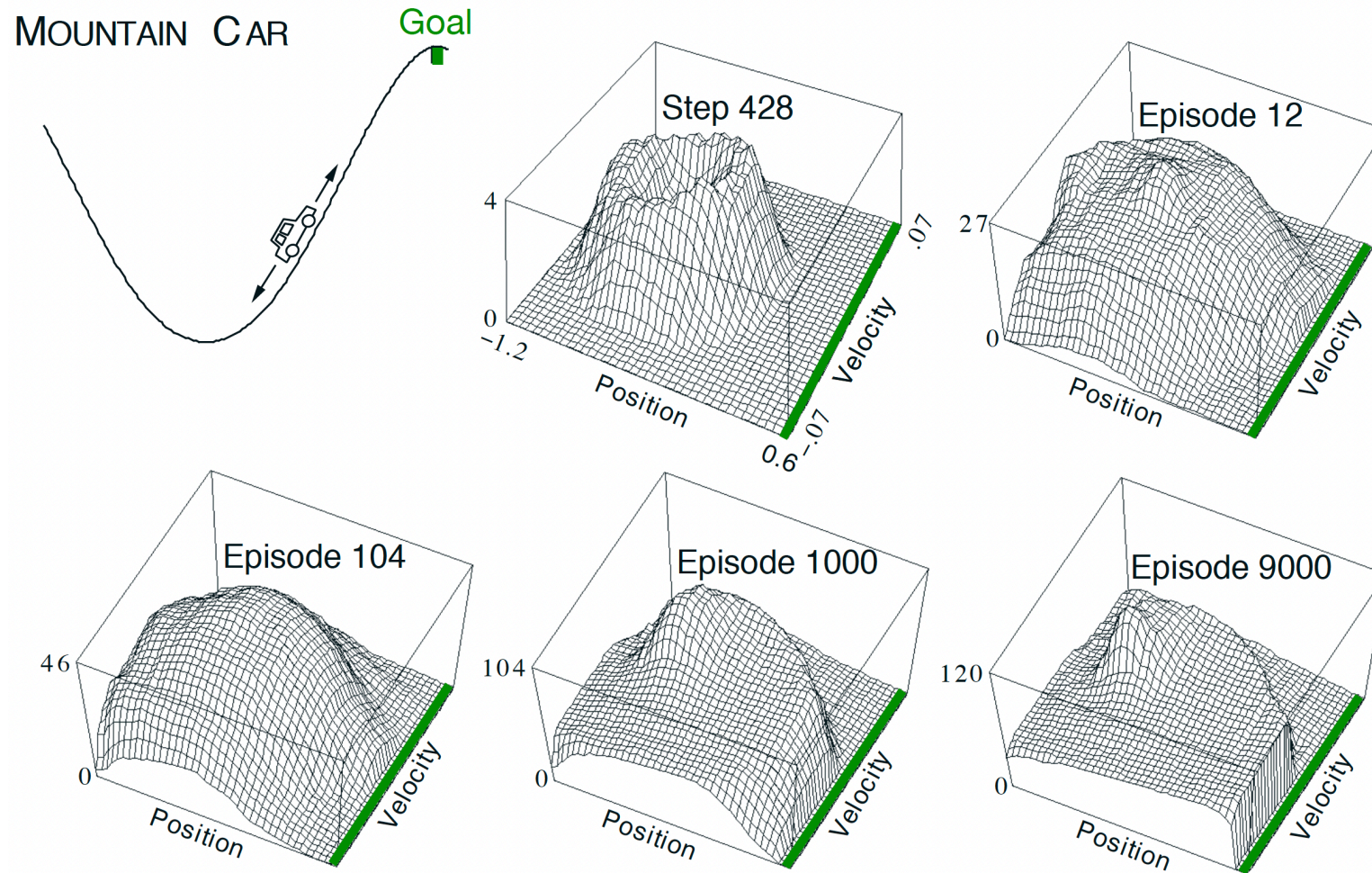
$$\psi(s, a) = \sum_{i=1}^d \theta_i x_i(s, a),$$

$$x_i(s, a) = \begin{cases} 1 & (s, a) \in \text{tile } \#i \\ 0 & \text{otherwise} \end{cases}.$$



Tile coding (Sutton and Barto 1998, Figure 9.9)

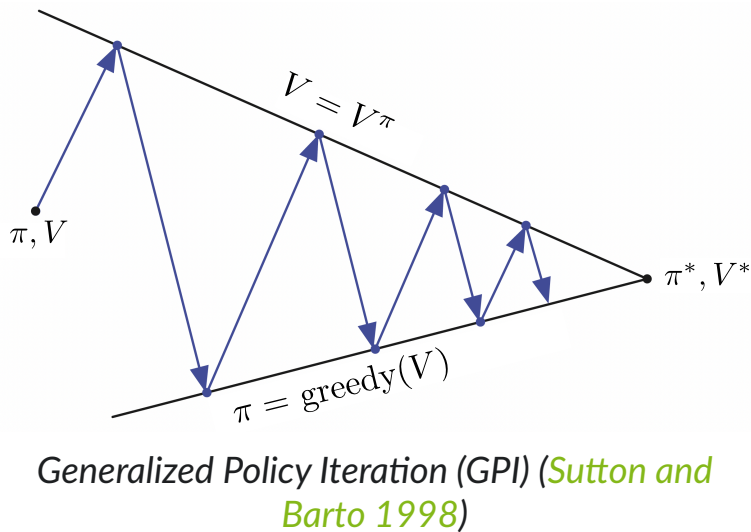
Example: Mountain car (2)



The Mountain Car task (upper left panel) and the cost-to-go function ($-\max_{a \in \mathcal{A}} Q_{\theta}(s, a,)$) learned during one run. (Sutton and Barto 1998, Figur 10.1)

Warning: a huge drawback of function approximation

- Recall tabular **policy improvement theorem**: guarantee to find a globally better or equally good policy in each update step.
- Parameter updates of the form (7) have an impact on $Q(s, a)$ for **all** s and a , not just the one we visited!



Loss of the policy improvement theorem

- When using function approximation, the policy improvement theorem does no longer apply.
- While improving the policy, we may impair it in other places *at the same time*.

Deep Q networks (DQN)

General strategy behind DQN

- Introduced in the famous DeepMind paper ([Mnih et al. 2015](#)).
- Same off-policy strategy as in the *classical/tabular* Q -learning algorithm:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a' \in \mathcal{A}} Q(s', a') - Q(s, a) \right].$$

- Now: incremental learning using semi-gradients (Eq. (7)) and TD(0) bootstrapping:

$$\theta \leftarrow \theta + \alpha \left[r + \gamma \max_{a' \in \mathcal{A}} Q_{\theta}(s', a') - Q_{\theta}(s, a) \right] \nabla_{\theta} Q_{\theta}(s, a). \quad (8)$$

But with two important distinctions

1. An **experience replay buffer** for batch learning replaces the single-sample updates.

The motivation behind this

1. **Sample efficiency**
 - *Experience replay*: efficiently use available data.
 - *Stabilize learning* by reducing the correlation between samples
 $\Rightarrow$ sampling from a replay buffer leads to i.i.d.-like data.
2. **Stabilization of learning**

2. A **separate set of weights** $\bar{\theta}$ for the bootstrapped Q -target ((8) is **not** a gradient).

- targets don't change in inner loop $\Rightarrow$ well-defined learning problem.

DQN working principle (1)

- Take actions u based on $Q_\theta(s, a)$ (e.g., ϵ -greedy).
- Store observed tuples (s, a, r, s') in memory buffer $\mathcal{D}$.
- Sample mini-batches $\mathcal{B}$ from $\mathcal{D}$.
- Calculate bootstrapped Q-target with a target network with weights $\bar{\theta}$:

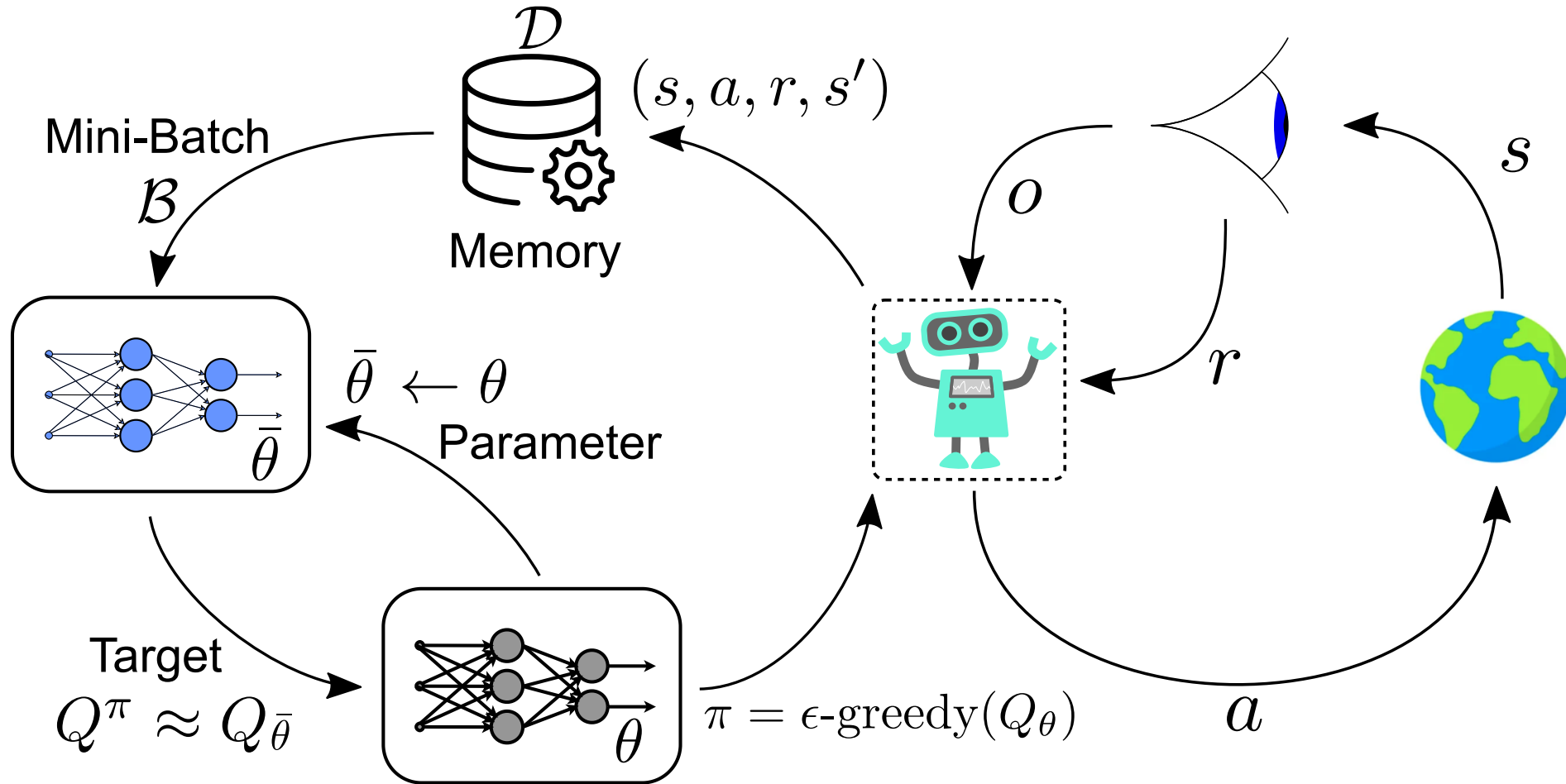
$$Q^\pi(s, a) \approx r + \max_{a \in \mathcal{A}} Q_{\bar{\theta}}(s, a).$$

- Optimize MSE loss between above targets and the regular approximation $Q_\theta(s, a)$ using $(s_i, a_i, r_i, s'_i) \in \mathcal{B}$:

$$L(\theta) = \sum_{i=1}^{|\mathcal{B}|} \left(\underbrace{r_i + \max_{a \in \mathcal{A}} Q_{\bar{\theta}}(s'_i, a)}_{=y_i \text{ (target)}} - Q_\theta(s_i, a_i) \right)^2,$$
$$\theta \leftarrow \theta + \alpha \sum_{i=1}^{|\mathcal{B}|} [y_i - Q_\theta(s_i, a_i)] \nabla_\theta Q_\theta(s_i, a_i).$$

- Update $\bar{\theta}$ based on θ from time to time.

DQN working principle (2)



Adapted from (Abdelwanis et al. 2026)

The DQN algorithm

Algorithm: The “classic” DQN algorithm (Mnih et al. 2015)

Input:

- a differentiable, parameter-dependent function $Q_\theta : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$
- learning rate α , greedy parameter ϵ , update rate $N_\theta \in \mathbb{N}$

Initialize: Q function weights $\theta = \bar{\theta} \in \mathbb{R}^d$ arbitrarily, $\mathcal{D} = \{\}$

for $k = 1, 2, \dots, K$ episodes:

Initialize s_0

for $t = 0, 1, \dots, T - 1$:

Obtain $a_t \sim \epsilon$ -greedy($Q_\theta(s_t, \cdot)$) and observe r_t and s_{t+1}

Store tuple (s_t, a_t, r_t, s_{t+1}) in $\mathcal{D}$

Sample minibatch $\mathcal{B}$ from $\mathcal{D}$ (after initial warm-up)

Compute targets for $(s_i, a_i, r_i, s'_i) \in \mathcal{B}$

$$y_i = r_i + \max_{a \in \mathcal{A}} Q_{\bar{\theta}}(s'_i, a) \quad (\text{or } y_i = r_i \text{ if } s'_i \text{ is terminal})$$

Update θ via gradient descent on $L(\theta)$ with learning rate α
if $t \bmod N_\theta = 0$ then $\bar{\theta} \leftarrow \theta$

Remarks:

1. For the standard MSE loss

$$L(\theta) = \sum_{i=1}^{|\mathcal{B}|} \left(y_i - Q_\theta(s_i, a_i) \right)^2,$$

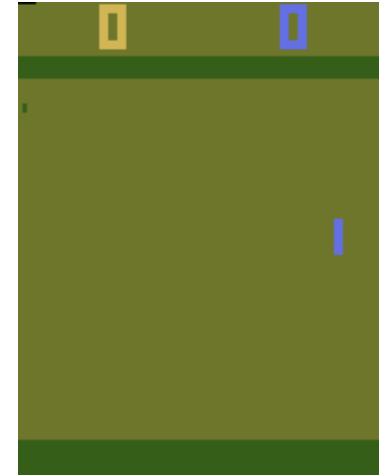
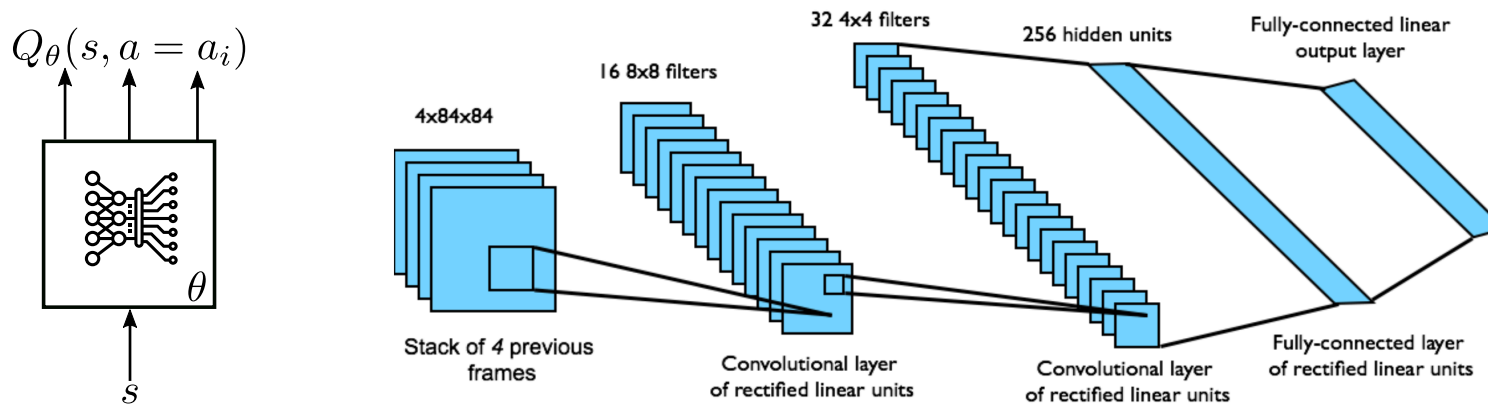
the weight update becomes the one we have seen before:

$$\theta \leftarrow \theta + \alpha \sum_{i=1}^{|\mathcal{B}|} [y_i - Q_\theta(s_i, a_i)] \nabla_\theta Q_\theta(s_i, a_i).$$

2. Apart from standard gradient descent, advanced (i.e., momentum-based) descent algorithms are possible (e.g., RMSProp, Adam, ...)

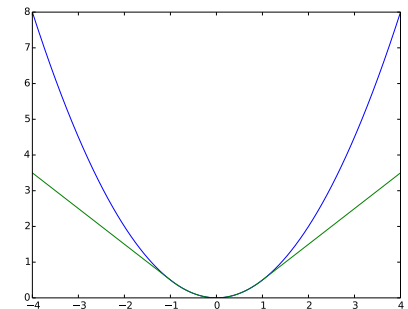
Example: Atari game Pong (1)

- *State s* :
 - 210×160 pixels with 3 channels each (RGB) and values $s_i \in \{0, \dots, 255\}$,
 - stacking of multiple frames,
 - some additional preprocessing.
- *Action a* : 18 actions in total, (6 buttons, some of which can be pushed at the same time).
- *Architecture for Q_θ* (Mnih et al. 2015):



Pong setup: More details in the [Farama Gymnasium](#) description.

- *Loss function $L(\theta)$* : Huber loss $L_\beta(\theta) = \begin{cases} \frac{1}{2}\theta^2, & \theta \leq \beta \\ \beta(|\theta| - \frac{1}{2}\theta), & \text{otherwise} \end{cases}$.
- *Training*: RMSProp descent algorithm.



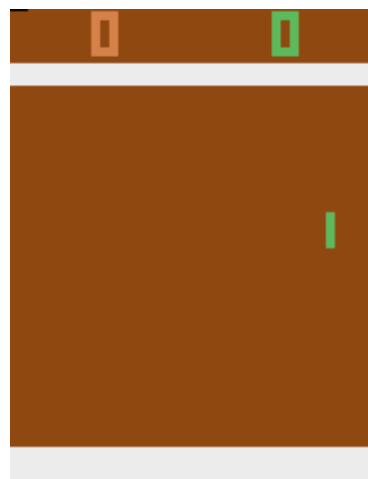
Huber loss ([Wikipedia](#))

- *Annealing* of the exploration rate: ϵ goes from 1 to 0.1 (or 0.05) over the first 10^6 frames.

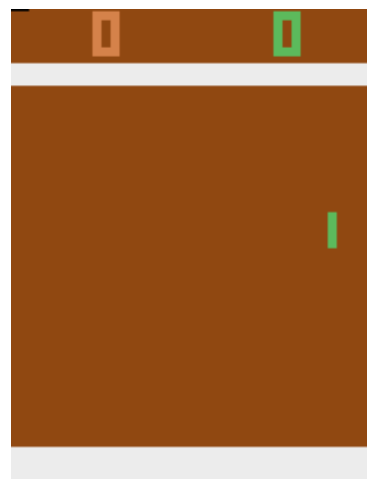
Example: Atari game Pong (2)



0 steps



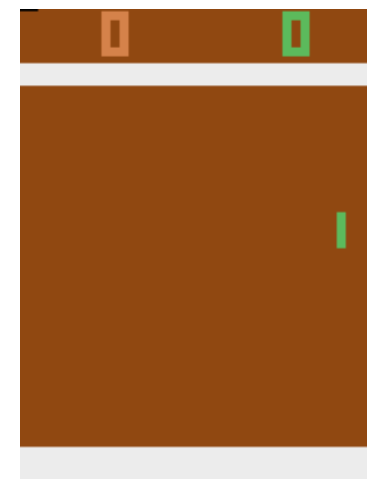
100,000 steps



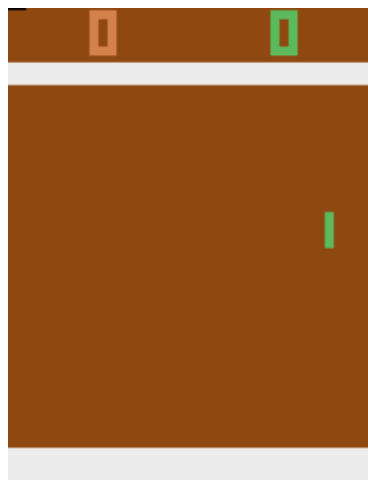
300,000 steps



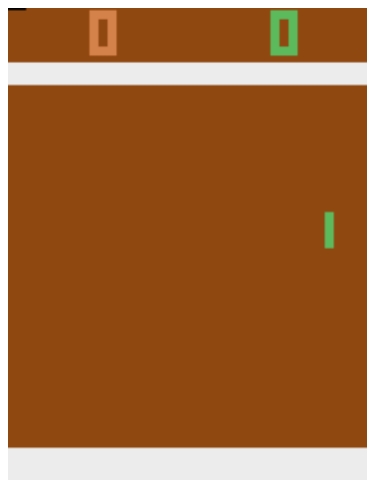
500,000 steps



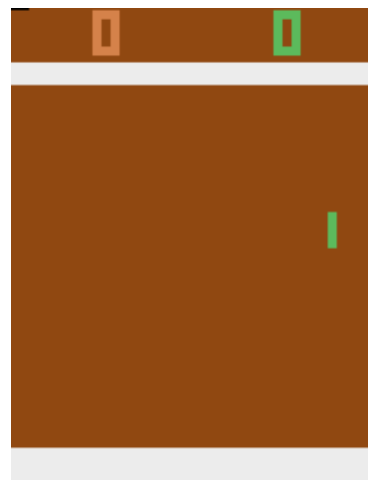
800,000 steps



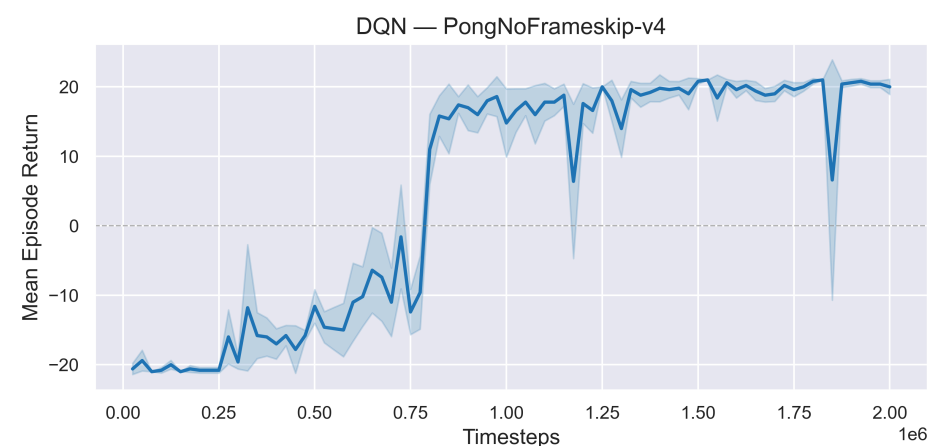
1,000,000 steps



1,500,000 steps

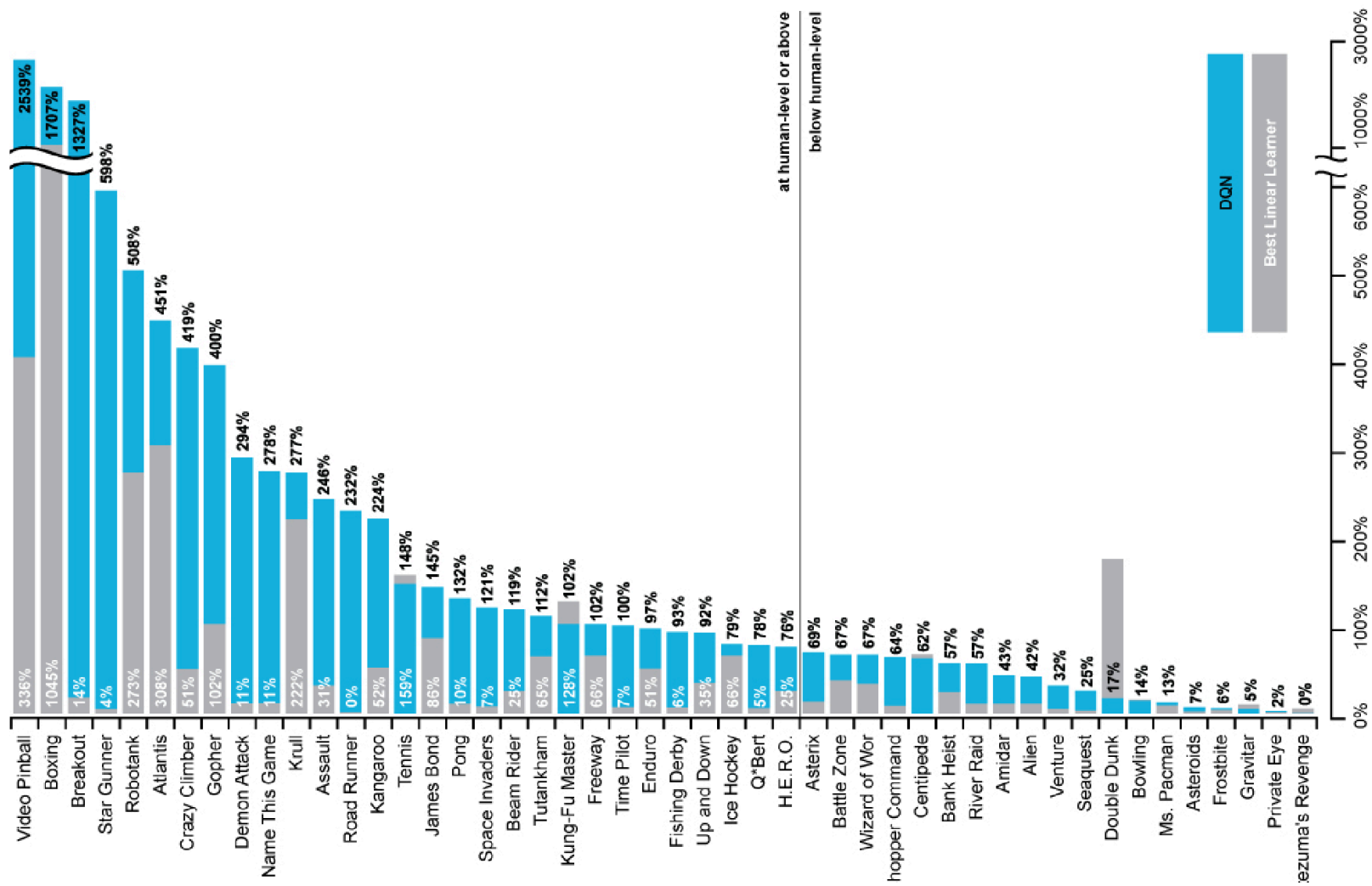


2,000,000 steps

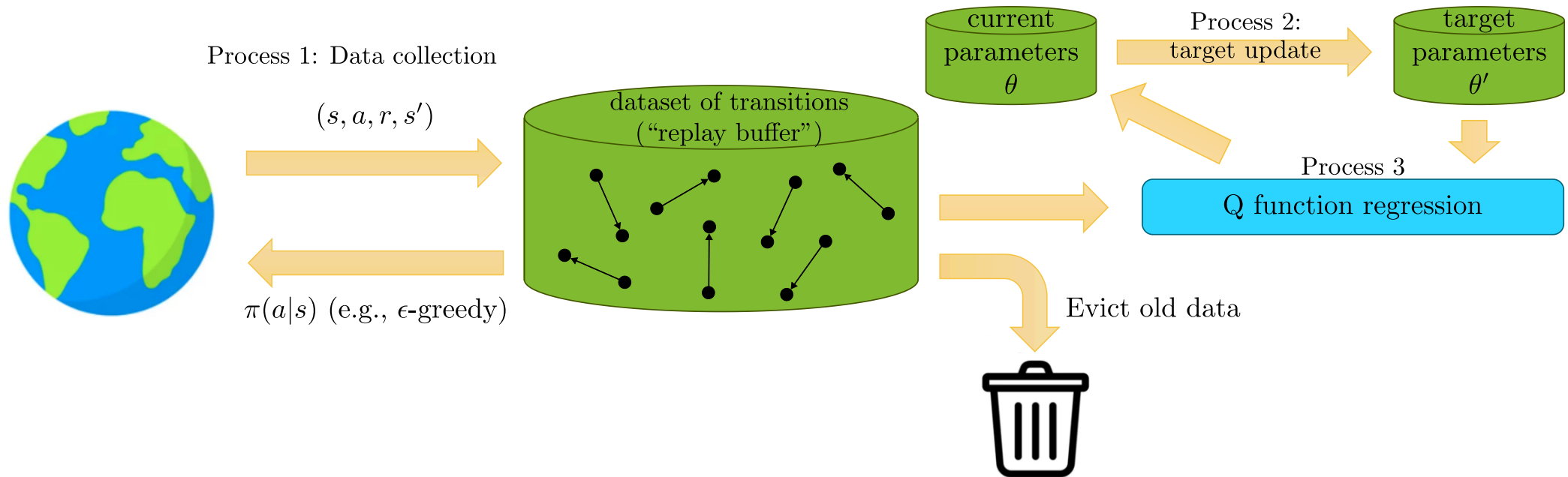


Return

Example: Atari games



The general view of Q -learning



Inspired by Sergey Levine's [CS285 lecture](#).

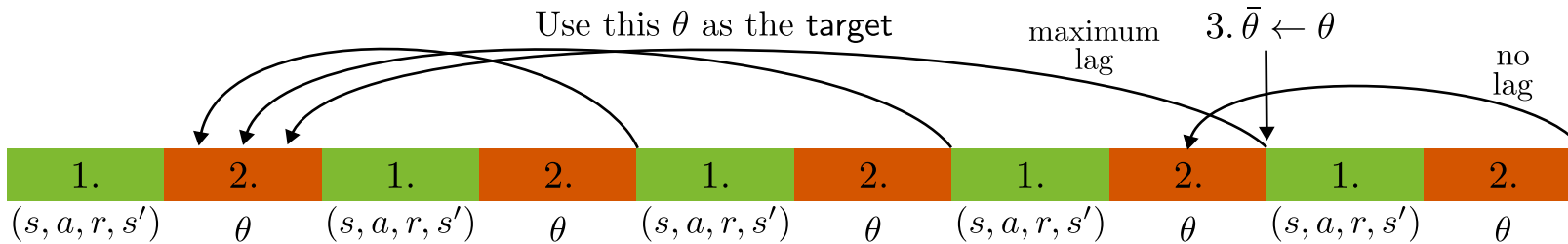
- **DQN:** Processes 1 and 3 run at the same speed, process 2 is slower.
- **Online Q -learning:** evict immediately, Process 1, 2 and 3 all run at the same speed.
 - Value or Q -learning as in the last lecture, followed directly by a policy improvement step.
 - Also similar to the value iteration algorithm from dynamic programming.

- Many variations: data collection and eviction strategies, frequencies and repetitions of the individual process steps, ...

DQN extensions & modifications

Alternative target networks (Polyak averaging)

- Consider the DQN algorithm and separate it into three main components:
 - Data collection, i.e., store tuples (s, a, r, s') in $\mathcal{D}$.
 - Improvement using mini batches $\mathcal{B}$ and a fixed target network, e.g., $\theta \leftarrow \theta + \alpha \sum_{i=1}^{|\mathcal{B}|} [y_i - Q_{\theta}(s_i, a_i)] \nabla_{\theta} Q_{\theta}(s_i, a_i)$.
 - Update target network every N_{θ} steps: $\bar{\theta} \leftarrow \theta$.
- Let's visualize these steps (with $N_{\theta} = 4$, even though it's usually much larger):



- The lag is different in different places! This doesn't need to be a problem, but it feels *uneven*.
- Alternative: **Polyak averaging**. In every step, perform the *linear interpolation*

$$\bar{\theta} \leftarrow \tau \bar{\theta} + (1 - \tau) \theta$$

- When initializing $\bar{\theta} = \theta$, this linear interpolation tends to work for nonlinear models (i.e., NNs) as well!

Double Q networks (Hasselt, Guez, and Silver 2016)

Remember the *maximization bias* and double Q-learning?
⇒ Don't use same network to choose the action and estimate the value!

Double deep Q-learning:

- Just use the **current network** (θ) to evaluate the action.
- Still use the **target network** ($\bar{\theta}$) to evaluate the value.

$$y = r + \gamma Q_{\bar{\theta}}(s', \arg \max_{a \in \mathcal{A}} Q_{\theta}(s', a))$$

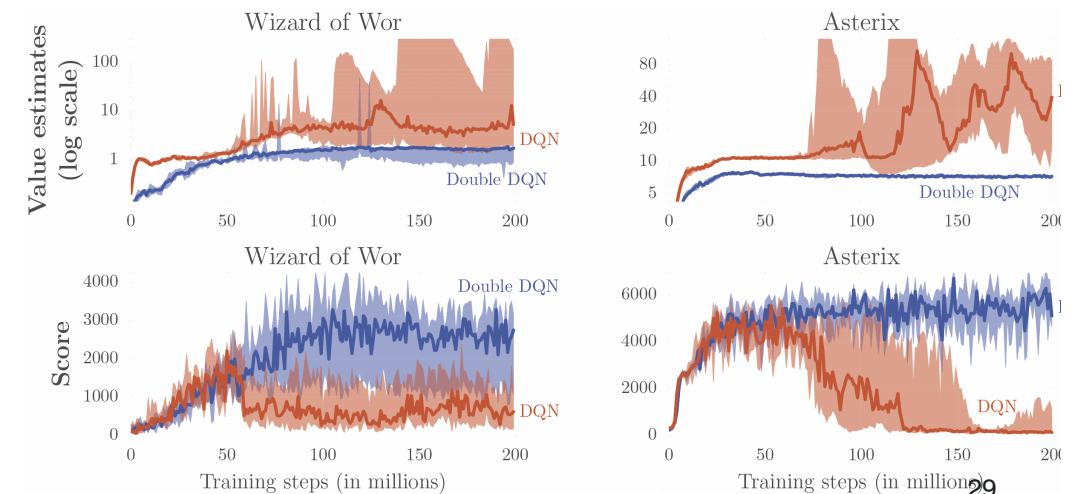
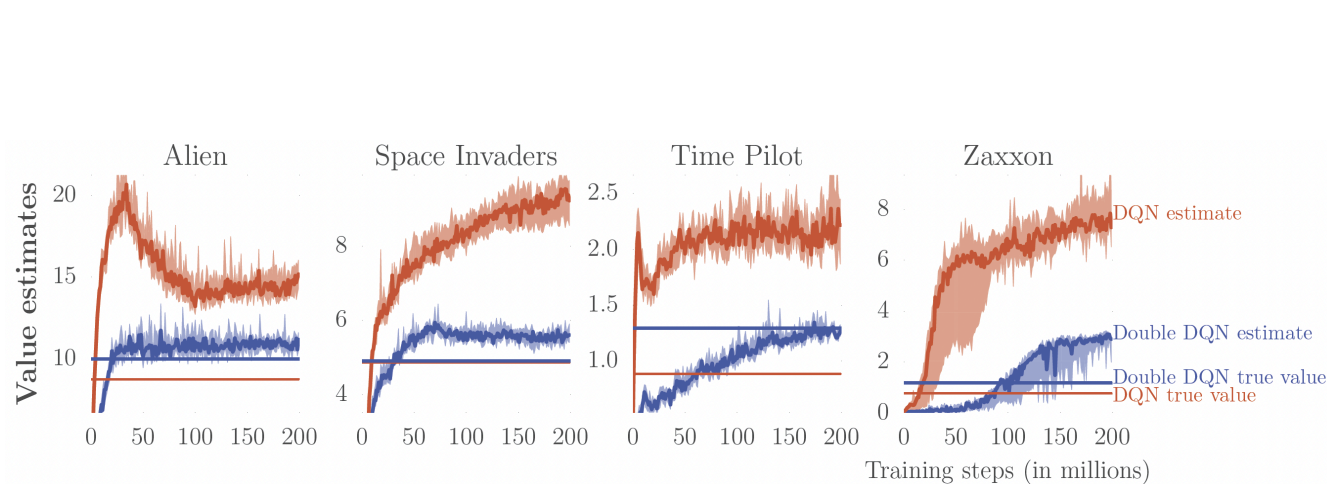
Algorithm recap: Tabular Q-Learning.

- Take a_t (ϵ -greedy on $Q_1 + Q_2$), observe (r_t, s_{t+1})
- if $\text{rand}() > 0.5$ then

$$y = r_t + \gamma Q_2(s_{t+1}, \arg \max_{a \in \mathcal{A}} Q_1(s_{t+1}, a))$$

$$Q_1(s_t, a_t) \leftarrow Q_1(s_t, a_t) + \alpha [y - Q_1(s_t, a_t)]$$

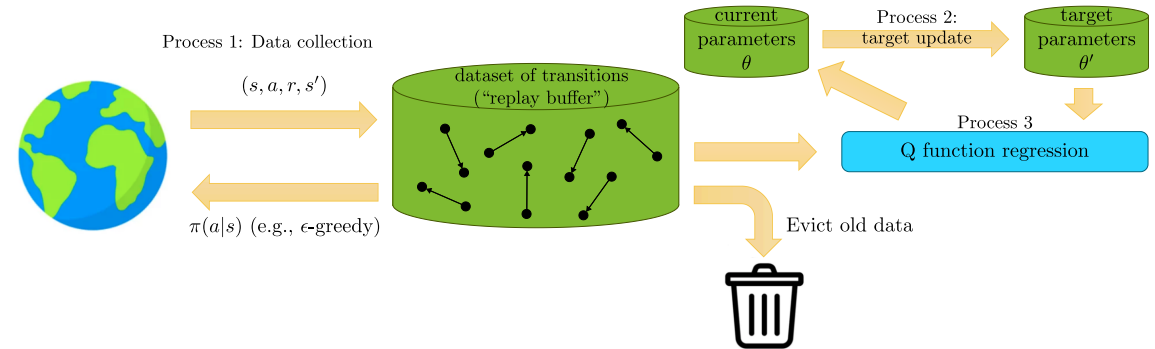
- else: ...



Prioritized experience replay (Schaul et al. 2016)

- Samples in the replay buffer are *not equally important*.
- From some examples, you can learn more than from others.
- What would be a good *criterion to assess the relevance* of a particular sample? $\Rightarrow$ The **TD error**

$$\delta_i = r_i + \max_{a \in \mathcal{A}} Q_{\bar{\theta}}(s'_i, a) - Q_{\theta}(s_i, a_i)$$



Inspired by Sergey Levine's CS285 lecture.

- Instead of sampling uniformly from the replay buffer, we assign an individual probability to each sample:

$$P(i) = \frac{p_i^\alpha}{\sum_{k=1}^{|\mathcal{D}|} p_k^\alpha}, \quad p_i = |\delta_i| + \epsilon.$$

- Here, $\alpha \geq 0$ determines the degree of prioritization ($\alpha = 0$ is the uniform case).
- The parameter $\epsilon > 0$ ensures that all samples are drawn with non-zero probability.

- A version that's less sensitive to outliers: rank the transitions according to $|\delta_i|$:

$$p_i = \frac{1}{\text{rank}(i)}.$$

- **But:** this introduces a bias (we sample from a different, uncontrollable distribution) $\Rightarrow$ Use importance sampling!

n -step returns

The concept of TD(n) can be extended to deep Q -learning in a straightforward fashion:

$$y_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots + \gamma^{n-1} r_{t+n-1} + \gamma^n \max_{a \in \mathcal{A}} Q_\theta(s_{t+n}, a).$$

- + less biased target values when estimates using Q_θ are inaccurate
- + typically faster learning, especially early on
- only actually correct when learning on policy

Some remarks on continuous actions

What's the problem with continuous actions in DQN?

1. The action selection: $\pi(a|s) = \begin{cases} 1, & a = \arg \max_{a' \in \mathcal{A}} Q_{\theta}(s, a') \\ 0, & \text{otherwise} \end{cases}$ (or some ϵ -greedy alternative).

2. The target calculation: $y_i = r_i + \max_{a \in \mathcal{A}} Q_{\bar{\theta}}(s'_i, a)$.

$\Rightarrow$ if $\mathcal{A}$ is continuous (i.e., $a \in \mathbb{R}^m$), then the maximization becomes a **challenging optimization** problem in itself!

What can we do?

1. Gradient based optimization (e.g., SGD) $\Rightarrow$ — This can be too slow!

2. Simple sampling:

$$\max_{a \in \mathcal{A}} Q(s, a) \approx \max \{Q(s, a_1), \dots, Q(s, a_p)\}, \quad \text{with } \{a_1, \dots, a_p\} \sim U(\mathcal{A}).$$

$\Rightarrow$ + Very efficient. + Easily parallelized. — Not very accurate.

3. Other stochastic optimization techniques $\Rightarrow$ — Usually also quite slow, or scale poorly.

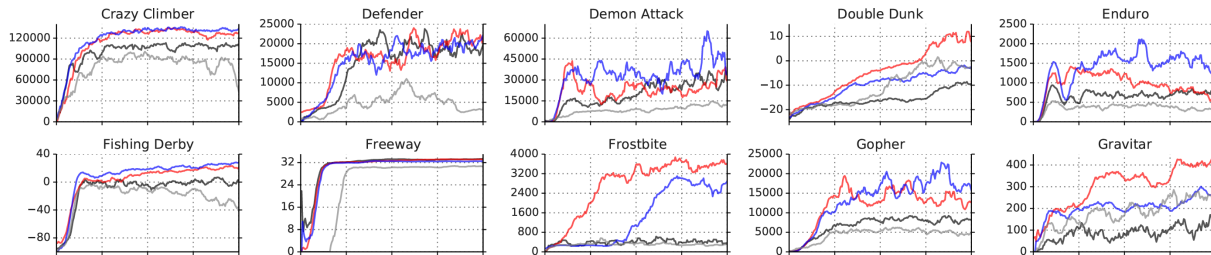
4. Train second network $\pi_\phi : \mathcal{S} \rightarrow \mathcal{A}$ whose output maximizes $Q_\theta(s, a)$:

$$Q(s, \pi_\phi(s)) \approx \max_{a \in \mathcal{A}} Q_\theta(s, a) \quad \Rightarrow \quad \text{Actor-critic methods (in two lectures)!}$$

Some practical tips

Following Sergey Levine's CS285 lecture:

- Q learning takes some care to stabilize.
 - Test on easy, reliable tasks first, make sure your implementation is correct.



(Schaul et al. 2016)

- Large replay buffers help improve stability.
- Looks more like fitted Q iteration.
- It takes time, be patient; performance might be no better than random for a while.
- Start with high exploration (ϵ) and gradually reduce.
- Bellman error gradients can be big; clip gradients or use Huber loss.
- Double Q learning helps a lot in practice: it's simple and has no downsides.
- n -step returns also help a lot, but have some downsides.
- Schedule exploration (high to low) and learning rates (high to low). The Adam optimizer can help too.
- Run multiple random seeds, it's very inconsistent between runs.

Least squares policy iteration

Linear action-value-function estimation

- Recall the LSTD algorithm from last lecture: batch least squares estimation $V^\pi(s)$ with linear models:

$$y = \begin{bmatrix} r_0 \\ \vdots \\ r_{T-1} \end{bmatrix}, \quad \Psi = \begin{bmatrix} \psi_0^\top - \gamma\psi_1^\top \\ \vdots \\ \psi_{T-1}^\top - \gamma\psi_T^\top \end{bmatrix}, \quad L(\theta) = \frac{1}{N} \|y - \Psi\theta\|_F^2, \quad \theta^* = (\Psi^\top \Psi)^{-1} \Psi^\top y = \Psi^\dagger y, \quad V_{\theta^*}(s) = \theta^{*\top} \psi(s).$$

- Let's do the same for Q values $\Rightarrow$ **LS-SARSA / LSTDQ**

$$Q_\theta(s, a) = \theta^\top \psi(s, a) = \sum_{i=1}^d \theta_i \psi_i(s, a) \quad \text{(Linear model)}$$

$$Q^\pi(s, a) \approx r + \gamma Q_\theta(s', a') \quad \text{(TD(0) bootstrapping)}$$

- Loss function:

$$L(\theta) = \sum_{t=0}^T \left(r_t - [\psi_t^\top - \gamma\psi_{t+1}^\top] \theta \right)^2 = \frac{1}{N} \|y - \Psi\theta\|_F^2, \quad \text{where } \psi_t = \psi(s_t, a_t).$$

- Same as before, only that the features depend on both states and actions: $\psi(s, a)$.

On-policy and off-policy LS-SARSA

- Same data collection procedure as before: batch mode $\Rightarrow \Psi$ and y .
- Regarding the data in Ψ , we can distinguish two cases:
 - the same policy π (*on-policy learning*),
 - an arbitrary policy $a' \sim \pi'(\cdot|s')$ (*off-policy learning*).
- If we apply off-policy LS-SARSA then
 - we retrieve the flexibility to collect training samples arbitrarily,
 - at the cost of an estimation bias based on the sampling distribution.
- Possible modifications:
 - To prevent numeric instability regularization is possible, e.g., ridge regression $\theta^* = (\Psi^\top \Psi + \epsilon I)^{-1} \Psi^\top y$
 - Recursive implementation for online usage is straightforward.

Least squares policy iteration (LSPI)

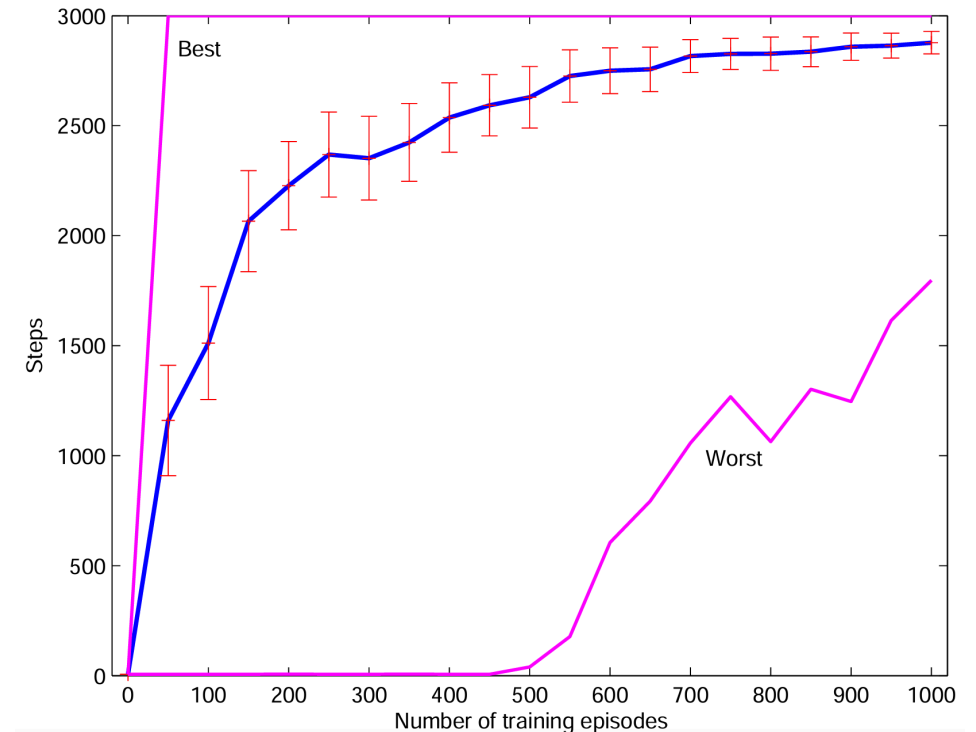
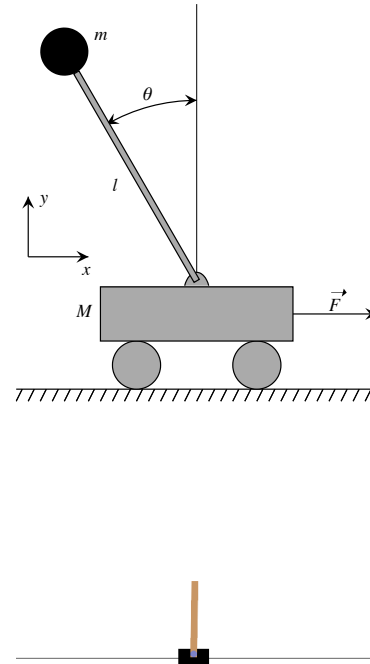
General idea:

- Apply general policy improvement (GPI) based on dataset $\mathcal{D}$,
- Policy evaluation by off-policy LS-SARSA,
- Policy improvement by greedy choices on predicted action values.

Some remarks:

- LSPI is an offline and off-policy control approach.
- Exploration is required by feeding suitable sampling distributions to $\mathcal{D}$:
 - ϵ -greedy choices based on Q_θ .
 - Completely random samples are conceivable as well.

Example: Pendulum on a cart ([Abdelwanis et al. 2026](#))



Balancing steps before episode termination with a clipping of maximum 3000 steps
([Lagoudakis and Parr 2003](#))

Summary / what you have learned

Summary / what you have learned

- The procedures from the approximate prediction can be transferred to value-based control in a straightforward way.
- Downside: the policy improvement theorem no longer applies in the approximate RL case.
 - Control algorithms may diverge.
 - Performance trade-offs between different parts of state-action space could emerge.
- Off-policy batch learning approaches allow for efficient data usage.
 - LSPI uses LS-SARSA on linear function approximation.
 - DQN extends Q -learning on non-linear approximation with additional tweaks (experience replay, target networks,...).
 - However, a prediction bias results (off-policy sampling distribution).

References

- Abdelwanis, Ali, Barnabas Haucke-Korber, Darius Jakobeit, Wilhelm Kirchgässner, Marvin Meyer, Maximilian Schenke, Hendrik Vater, Oliver Wallscheid, and Daniel Weber. 2026. "Reinforcement Learning: A Comprehensive Open-Source Course." *Journal of Open Source Education* 9 (97). The Open Journal: 306. doi:[10.21105/jose.00306](https://doi.org/10.21105/jose.00306).
- Hasselt, Hado van, Arthur Guez, and David Silver. 2016. "Deep Reinforcement Learning with Double q-Learning." *Proceedings of the AAAI Conference on Artificial Intelligence* 30 (1).
- Lagoudakis, Michail G, and Ronald Parr. 2003. "Least-Squares Policy Iteration." *Journal of Machine Learning Research* 4: 1107–49.
- Mnih, Volodymyr, Koray Kavukcuoglu, David Silver, Andrei A. Rusu, Joel Veness, Marc G. Bellemare, Alex Graves, et al. 2015. "Human-Level Control Through Deep Reinforcement Learning." *Nature* 518 (7540): 529–33.
- Schaul, Tom, John Quan, Ioannis Antonoglou, and David Silver. 2016. "Prioritized Experience Replay." In *International Conference on Learning Representations (ICLR)*.
- Sutton, R. S., and A. G. Barto. 1998. *Reinforcement Learning: An Introduction*. MIT Press.